Ricardo Costa

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ABOUT

Software engineer focused on developer tools, software verification, and agentic AI workflows. Working on agentic code review with Gitar at Sonar, alongside research on LiquidJava, a static analysis tool for Java. Also experienced in backend systems, web and mobile applications, and cloud computing.

EXPERIENCE

Sonar 2026 – Present
Software Engineer Geneva, Switzerland

Agentic Code Review (Gitar)

LASIGE Research Unit 2025 – 2026 (Expected)
Software Engineering Researcher Lisbon, Portugal

Project: Improving the usability of LiquidJava, a liquid type system for Java, through enhanced verification feedback, extended IDE support, and a better overall developer experience.

- **Verification Feedback:** Redesigned diagnostics and implemented traceable predicate simplification, counterexamples, and actionable hints to make verification failures easier to understand and resolve.
- **IDE Integration:** Enhanced the VS Code extension with syntax highlighting, autocomplete, hover information, and interactive views for exploring diagnostics, inspecting the verification context, and visualizing typestate protocols.
- **Verifier Improvements:** Improved verification robustness and predictability through multiple error reporting, complete project verification, and closer alignment with Java syntax and semantics.

Contributions: 120+ merged pull requests to the verifier and 80+ to its VS Code extension.
Funding: FCT | LASIGE

EDUCATION

Faculdade de Ciências da Universidade de Lisboa (FCUL) 2024 – 2026 (Expected)
MSc in Computer Science and Engineering (Specialization in Software Engineering) Lisbon, Portugal

Current Grade: 17/20
Thesis: "Improving the Usability of LiquidJava"
Main Courses: Software Design (17/20), Compilation Techniques (18/20), Software Verification and Validation (16/20), Cloud Computing (18/20), Software Security (17/20), Mobile Computing (19/20)

Instituto Superior de Engenharia de Lisboa (ISEL) 2021 – 2024
BSc in Computer Science and Engineering Lisbon, Portugal

Final Grade: 15/20
Final Project: "NoteSpace: Multi-Platform Web Application for Real-Time Document Collaboration & Sharing" (18/20)
Main Courses: Programming (19/20), Software Laboratory (20/20), Algorithms and Data Structures (16/20), Web Programming (17/20), Mobile Programming (16/20)

PROJECTS

liquidjava | Improved verification feedback and developer experience

vscode-liquidjava | Extended IDE support for reading, writing and debugging refinements in VS Code

liquidjava-mcp | Java MCP server that exposes LiquidJava verification tools to LLM agents over stdio

NoteSpace | Real-time collaborative editor with markdown and version control support

AI Code Reviewer | Lightweight AI code review workflow powered by Cloudflare Workers AI and KV

Barista | LLM agent workflow for synthesizing LiquidJava specifications from Javadoc documentation

Threadit | REST API with microservices architecture, gRPC service communication, and GCP deployment

aguda-rs | Compiler for a custom programming language implemented in Rust

passman | Secure multi-user CLI password manager with local encrypted storage built in Rust

Sketch.io | Multiplayer drawing and guessing game using React, Node.js, and Socket.IO

AdventOfCode | Solutions for Advent of Code coding puzzles using Python (88 completed)

SKILLS

Programming Languages | Python, Java, JavaScript, TypeScript, Kotlin, Rust, Go, C, C#
Backend | REST, gRPC, Microservices, WebSockets, Spring Boot, Node.js, Express, Socket.IO, Maven, Gradle
Cloud & DevOps | GCP, Cloudflare Workers, Docker, Kubernetes, Git, GitHub Actions, CI/CD, Linux
Frontend | React, HTML, CSS, Vite, Webpack, PWA
Mobile | Android, Jetpack Compose
Databases | PostgreSQL, Firestore, MongoDB, Cloudflare KV, Elasticsearch
Compilers & Static Analysis | Parsers, Type Systems, SMT Solvers, LLVM
Developer Tooling | VS Code Extension API, LSP, JSON-RPC
AI Engineering | AI Agents, Agentic Workflows, Cloudflare Workers AI, MCP

PUBLICATIONS

"Barista: Synthesizing Typestate Specifications with LLM Agents" | Catarina Gamboa, Paulo Canelas, Ricardo Costa, Márcio Caetano, Jonathan Aldrich, Alcides Fonseca (Accepted at Automated Software Engineering (ASE) 2026)

POSTERS

"Improving the Usability of LiquidJava" | Ricardo Costa, Catarina Gamboa, Alcides Fonseca

"Barista: Synthesizing Typestate Specifications with LLM Agents" | Catarina Gamboa, Paulo Canelas, Ricardo Costa, Márcio Caetano, Jonathan Aldrich, Alcides Fonseca (Best PhD Student Poster Award)

EVENTS

Poster Presenter | CIÊNCIAS Research & Innovation Day 2026 · "Improving the Usability of LiquidJava"
Speaker | JNation 2026 · "LiquidJava: Catch Bugs at Compile Time with Liquid Types"
Poster Presenter | 11th LASIGE Workshop · "Improving the Usability of LiquidJava"
Volunteer | Lisbon AI 2026